

Exp. 1	SETUP A PRIVATE CLOUD BY PERFORMING THE PROCEDURE USING A SINGLE NODE OPENSTACK IMPLEMENTATION
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AIM

To set up a Private Cloud using a single-node OpenStack implementation and create virtual machine instances.

OBJECTIVE

- To understand the concept of Private Cloud Computing.
- To install and configure OpenStack using DevStack.
- To create and manage virtual machine instances in a cloud environment.
- To understand cloud resource management using the OpenStack Dashboard.

SOFTWARE REQUIREMENTS

- Ubuntu Server 22.04 LTS
- OpenStack (DevStack)
- Virtual Machine Software (VirtualBox/VMware)
- Internet Connection
- Web Browser

THEORY

Cloud Computing is a technology that provides computing resources such as servers, storage, databases, networking, and software over the internet. A Private Cloud is a cloud infrastructure dedicated to a single organization and offers greater control, security, and customization.

OpenStack is an open-source cloud computing platform used for building and managing private and public cloud environments. It provides Infrastructure as a Service (IaaS) through a collection of services for computing, networking, storage, and identity management.

The major services of OpenStack include:

- Nova – Compute Service
- Neutron – Networking Service
- Glance – Image Service
- Keystone – Identity Service
- Cinder – Block Storage Service
- Swift – Object Storage Service

ARCHITECTURE

OpenStack Private Cloud Architecture:

User → Horizon Dashboard → Keystone Authentication → Nova Compute → Neutron Networking → Glance Images → Cinder Storage

PROCEDURE

STEP 1: INSTALL UBUNTU SERVER

1. Create a virtual machine using VirtualBox or VMware.
2. Allocate at least 4 GB RAM and 2 CPU cores.
3. Install Ubuntu Server.

STEP 2: UPDATE SYSTEM PACKAGES

Execute the following commands:

```
sudo apt update
sudo apt upgrade -y
```

STEP 3: INSTALL GIT

Install Git for downloading DevStack:

```
sudo apt install git -y
```

STEP 4: DOWNLOAD DEVSTACK

Clone the DevStack repository:

```
git clone https://opendev.org/openstack/devstack
```

STEP 5: CONFIGURE DEVSTACK

Navigate to the DevStack directory:

```
cd devstack
```

Create a configuration file named local.conf and provide the required credentials.

STEP 6: INSTALL OPENSTACK

Run the installation script:

```
./stack.sh
```

Wait for the installation process to complete successfully.

STEP 7: ACCESS OPENSTACK

DASHBOARD

Open a web browser and enter:

```
http://<server-ip>/dashboard
```

Login using the administrator credentials.

STEP 8: CREATE PROJECT AND USER

1. Open Identity → Projects.
2. Create a new project.
3. Create users.
4. Assign appropriate roles.

STEP 9: LAUNCH VIRTUAL MACHINE INSTANCE

1. Open Compute → Instances.
2. Click Launch Instance.
3. Select image, flavor, and network.
4. Launch the instance.

STEP 10: VERIFY CONNECTIVITY

1. Assign a floating IP address.
2. Connect to the instance using SSH.
3. Verify connectivity using ping commands.

Commands Used

```
sudo apt update
sudo apt upgrade -y
sudo apt install git -y
git clone https://opendev.org/openstack/devstack
cd devstack
./stack.sh
openstack server list
openstack network list
openstack image list
```

OBSERVATIONS

Parameter	Status
Ubuntu Installation	Successful
DevStack Installation	Successful
OpenStack Dashboard Access	Successful
Project Creation	Successful
VM Instance Creation	Successful
Network Connectivity	Verified

EXPECTED OUTPUT

- OpenStack Dashboard accessible through a web browser.
- Project, users, and roles successfully created.
- Virtual machine instances launched successfully.
- Network connectivity verified.

RESULT

A Private Cloud environment was successfully created using OpenStack. Virtual machine instances were launched and managed through the OpenStack Dashboar